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Automatic hyperbola detection and fitting in GPR B-scan image☆

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ABSTRACT

Detecting buried objects from ground penetrating radar (GPR) profiles often requires manual interaction and plenty of time. This paper presents an automatic scheme for buried objects detection and localization. First, a trained deep learning framework — Faster R-CNN with data augmentation strategy is applied to identify hyperbolic signatures from a gray GPR B-scan image, which is capable of not only recognizing whether a B-scan profile contains traces of buried object, but also detecting candidate hyperbola region. Then, the detected rectangle region is extracted and transformed to a binary image, a novel double cluster seeking estimate (DCSE) algorithm is proposed to separate object point cluster from each other and enable the identification of hyperbolic signatures. Subsequently, a column-based transverse filter points (CTFP) method is utilized to extract hyperbola fitting points automatically from the validated point cluster. Downward opening hyperbola fitting is carried out and their respective peaks are obtained finally. The proposed scheme is able to extract information from GPR B-scan images automatically and efficiently; it is validated significant performance in the analysis of synthetic and on-site GPR data sets.

1. Introduction

As a non-destructive detection tool, Ground Penetrating Radar (GPR) has been widely used to detect unknown objects in shallow subsurface [1]. Military issues include detection of mines and buried explosive hazards (BEHs) [2–6]. In the field of civil engineering, advanced GPR systems are increasingly used for road pavement monitoring [7–9]. GPR has been recently proposed as a method to estimate moisture content in soils. More in depth, one of the most relevant causes of road pavement damage is often referable to water or clay intrusion in structural layers [10]. In bridge surveys, GPR is used to measure concrete thickness, estimate relative water moisture content variation and mark or map rebar locations [11,12]. Engineering applications mainly involve non-destructive testing of structures and utility lines. There is an urgent need to devise a reliable method for collecting utility location data, and to model and visualize the associated positional uncertainties to prevent utility strikes [13–15].

GPR utilizes underground materials' the different electromagnetic properties to detect underground region by emitting high frequency electromagnetic waves into the ground and receiving return signal. Moreover, GPR B-scan profile of underground target presents image feature with a pair of hyperbolic-shaped signatures. In this way, the detection of underground target can be transformed into the extraction of hyperbola. In practical applications, GPR images are often noisy due to hardware modules noise, the heterogeneity of subsurface medium, and mutual wave interactions, and thus, it is challenging to automatically cope with GPR images and extract hyperbolae from them. Considerable research has been devoted in this area and many different strategies have been employed to tackle this topic as follows.

Over the past decade, Hough transform [16] techniques are used successfully for deformed shapes fitting. By introducing a weighting factor to Hough transform, Borgioli et al. [17] solved the problem of hyperbolae overlap when the pipes are close together. Windsor et al. [18] recorded an associative sets of position/time data pairs in the generalized Hough transform method, which coordinates with a conventional least-squares (LS) algorithm to explain certain position parameters of targets. In common, these methods require model specifications in advance and large computation costs, which limit further

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applications to practical situations. Unlike the Hough transform, quadratic curves could be searched and distinguished through the LS method for GPR images. The LS methods in [19–21] are especially designed for hyperbola fitting, but most of them could only recognize one curve in an image and are not applicable to multi-curve detection because of the lack of image segmentations step. In practical cases, noises could not be completely removed, and an image might contain multiple hyperbolae.

Recently, machine learning-based methods also are applied to GPR images, which avoids template matching and narrows down the searching regions including hyperbola. In [22], a method is utilized to classify noise signals by neural nets during detecting process. Maas and Schmalzl [23] used the Viola-Jones (VJ) algorithm [24] to extract the target regions in GPR data. For these works, most application features need to be identified by experts, and the classification results depend on the quality of feature, which is getting harder as the volume of data increases.

Several publications have proposed detection algorithms based on Convolutional Neural Network (CNN) [25] for the analysis of GPR data. The CNN skips the traditional "feature engineering" step, which often associated with machine learning, and instead learn the feature representations of buried objects directly from GPR B-scan images as done in [2–6,26]. In 2015, Lance E. et al. [2] applied CNN to extract meaningful signatures from GPR B-scans and classify BEHs. Several heuristics are used to prevent over-training, including cross validation, network weight regularization, and "dropout". Based on the previous research, Lance E. et al. [3] added an extra skill — data augmentation (DA) in 2016, which is a widely-applied technique to increase volume and variability of available training data. In [4,5], Daniel R. et al. investigated the initialization step of pretraining CNN to solve few labeled samples of GPR data for target detection. While CNN-based methods for hyperbola detection have been improved rapidly in the last two years, the size and quantity of the input GPR on-site images are often limited, most approaches only implement classification steps, lacking the target extracting and curve fitting methods.

Newly, there are some works that implement an integrated system, which is able to detect and fit hyperbola from GPR images automatically. In [27], column-connection clustering (C3) algorithm was proposed to separate the hyperbolae with intersections, further, the outputs from C3 are again classified by a neural-network-based method. In this model, the C3 algorithm scans each pixel in GPR B-scan image horizontally for clustering. However, in GPR B-scan images, the hyperbola is vertically downward opening, this important feature is not considered in this C3 algorithm. Zhou et al. [28] proposed open-scan clustering algorithm (OSCA) to solve this problem. The OSCA scans preprocessed binary image longitudinally and conducts clustering not only by the connections of pixels but also the opening information. So, the OSCA could identify the point clusters with downward-opening signatures. However, the OSCA conducted on a whole image is not suitable, because it is hard to handle large on-site data set which contains too much non-stationary noise and thus complicates follow-up processing. In addition, both the above two methods are not explained how to automatically obtain fitting points before performing hyperbola fitting algorithm.

In order to solve above issues, an automatic scheme is proposed to localize hyperbola targets in GPR B-scan images; its processing flowchart is shown in Fig. 1, which could be divided into four parts. In this paper, we investigate applying the Faster Region-based Convolutional Neural Network (Faster RCNN) [29] to GPR B-scan hyperbola detection. Faster R-CNN could generate rectangular regions, and then following processes will be operated on a few object regions rather than a whole image, which narrows down the processed region to obtain hyperbolic point clusters easily. In addition, a DA strategy is also adopted to increase volume of training set for Faster R-CNN. Moreover, Faster R-CNN has recently demonstrated impressive results on specific applications and datasets for object detection, such as vehicle detection [30] and

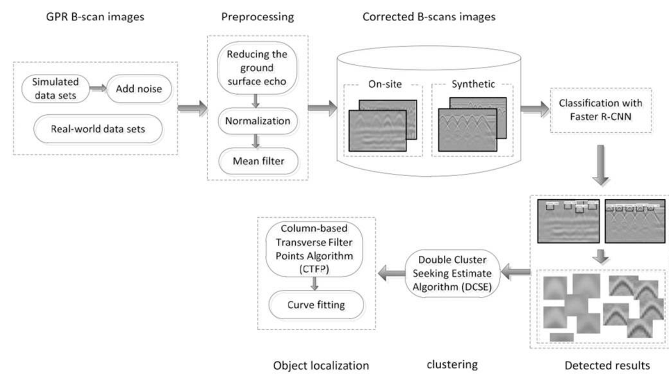

Fig. 1. The automated target identification scheme in the GPR B-scan image.

face detection [31,32]. So, it's feasible to apply Faster R-CNN to GPR B-scan images. Then, the detected rectangle region is extracted and transformed to a binary image, and the discrete noisy points are removed by opening and closing operations. Subsequently, a novel double cluster seeking estimate (DCSE) algorithm scans the binary image twice to identify hyperbolic signatures. Finally, a column-based transverse filter points (CTFP) method is utilized to extract fitting points automatically from the validated point cluster, followed by the hyperbola fitting algorithm that is applied to fit the identified point clusters and obtain accurate estimations of the objects.

The rest of this paper is organized as follows. The review and application of Faster R-CNN to hyperbola detection are presented in Section 2. The DCSE method is presented in Section 3, which is followed by CTFP method to extract fitting points automatically and hyperbola fitting in Section 4. The experimental results are shown and analyzed in Section 5, and finally, conclusions are drawn in Section 6.

2. Hyperbola detection via deep Faster R-CNN

In this section, we take a closer look at the Faster R-CNN approach as it applies to hyperbola detection. Before applying the proposed detection method, a series of preprocessing techniques are employed on GPR images. High amplitude region (white region) occurring at top rows of the image is the point at which the radiated radio frequency waves are reflected from the air-ground interface [Fig. 2(a)]. The first column value is subtracted from each column to eliminate high amplitude region, and a moving average filter processing is followed to reduce noise. An example image after the preprocessing is visualized in Fig. 2(b). The window size of the filter should not be too large but within a certain range, while the experimental results are not sensitive to it. A filter window size of 5×5 is chosen in our experiments.

2.1. Overview of Faster R-CNN

Recent advances in object detection have been driven by the success of CNN. Girshick et al. [33] introduced a region-based CNN (R-CNN) for object detection. First, a set of category-independent object proposals are generated by using selective search [34]. Second, the image region within each proposal is warped to a feature vector, and then fed into a classifier and also into a regressor. The R-CNN, however, requires a forward pass through the convolutional network for each object proposal in order to extract features, leading to a heavy computational burden. To mitigate this problem, the SPPnet [35] and Fast R-CNN [36] have been proposed to run through the CNN exactly once for the entire input image. Both the R-CNN and Fast R-CNN rely on the input generic object proposals, which usually come from a hand-crafted model such as selective search or EdgeBox [37].

To reduce the computational burden of proposal generation from the hand-crafted model, Faster R-CNN [29] was proposed for generic object detection, including two parts: (1) a Region Proposal Network

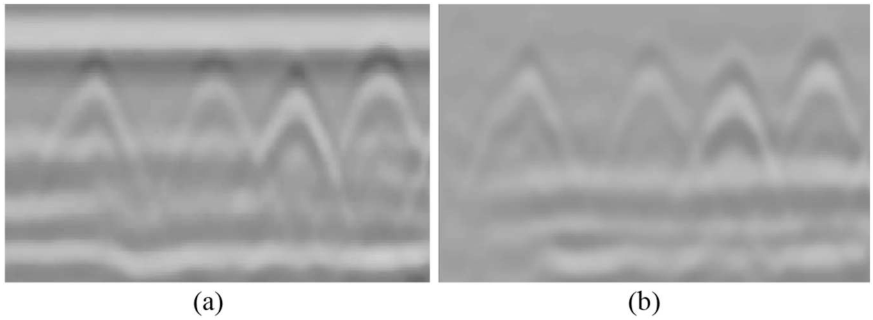

Fig. 2. Preprocessing results. (a) Input image. (b) Image after preprocessing.

(RPN) for generating a list of region proposals which likely contain objects; (2) a Fast R-CNN network for classifying a region into object and refining the boundaries of those regions. The two parts share common parameters in the convolution layers used for feature extraction, allowing this architecture to accomplish object detection tasks, in addition, it is possible to use a very deep network to generate high-quality object proposals.

2.2. Application of Faster R-CNN to hyperbola detection

Our scheme follows the deep learning framework of Faster R-CNN to recognize hyperbola signature on GPR B-scan data as shown in Fig. 3.

Due to the lack of on-site B-scan data for training CNN model, we propose to incorporate more synthetic data using gprMax toolbox [38]. Various configurations are considered where the objects with different sizes and materials are placed at different positions and depths. In addition, we also adopt the DA strategy to increase the amount of training data, which initially has a total of 838 images, including synthetic data set (748 images) and on-site data set (90 images). After applying DA, we obtain a total of 5866 images composed of 5236 synthetic images and 630 on-site images (details in the Section 5). In particular, we adopt DA strategy which includes horizontal and vertical stretching and compression, horizontal flipping, and image enhancement. The DA techniques have greatly increased the volume and variety of the training data and make the algorithms much less susceptible to minutia in the somewhat limited datasets.

An application of Faster R-CNN framework to the B-scan data sets consists of two main stages: 1) pre-train the ResNet-50 [39] on the ImageNet database; 2) train and fine-tune the Faster-RCNN (based on pre-trained CNN weights) using both on-site and synthetic GPR data after DA strategy. First, the ResNet-50 model is adopted to pre-train on the ImageNet data set, and the obtained pre-training model parameters are used for initializing the CNN model of Faster R-CNN. Next, we fine tune weights of the CNN model using own B-scan image data sets. The input image is fed into the trained Faster R-CNN, then the output image is obtained with bounding boxes and confidences as shown in Fig. 4(a): (1) Bounding box that contains expected hyperbola target; (2) Confidence that represents how likely a bounding box includes some

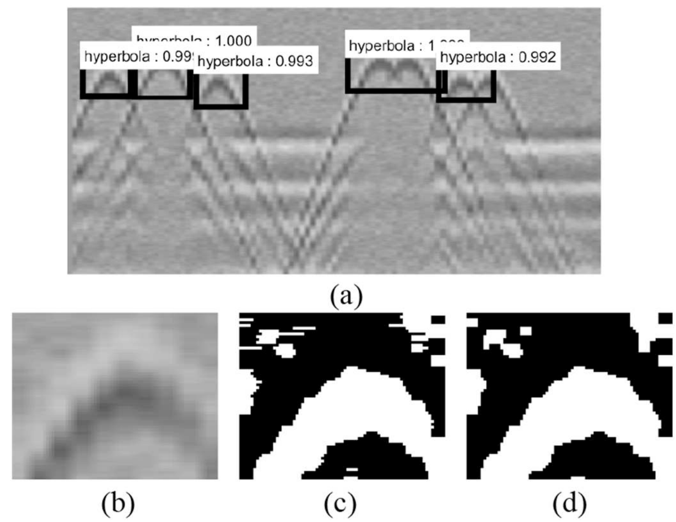

Fig. 4. (a) The output results of Faster R-CNN with bounding boxes and confidences. (b) An example of the detected target region. (c) The binary image after Otsu. (d) The processed binary image after closing and opening operations.

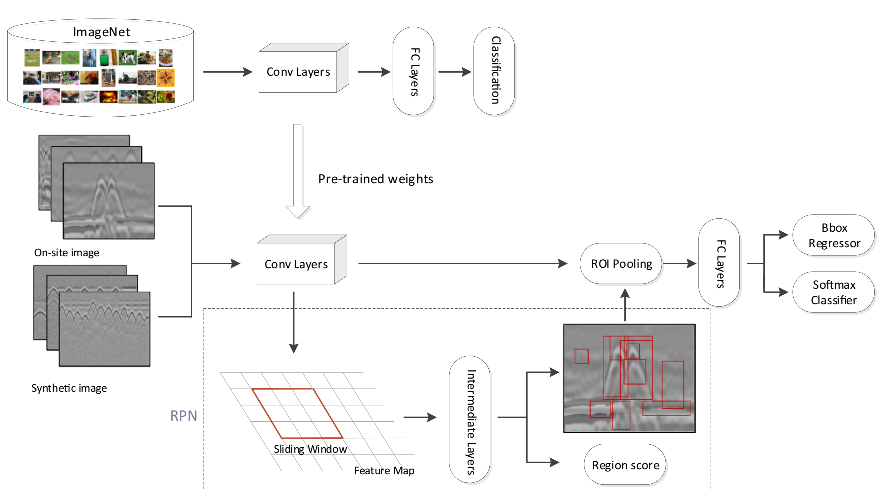

Fig. 3. The framework of Faster R-CNN for B-scan target detection.

hyperbolic signatures or not. The trained Faster R-CNN can be applied to classify the B-scan test images. Judged by the experimental results below in Section 5, Faster R-CNN works well for most gray hyperbolic signatures.

3. Double cluster seeking estimate algorithm

In this section, the follow-up processing method for detected region is carried first, and then the proposed DCSE algorithm is detailed subsequently.

3.1. Follow-up processing method for detected region

For the detected results in Section 2 Subsection 2.2, all rectangular regions are automatically extracted for further processing. By applying the Otsu [40] thresholding method on detected regions, the extracted target region I [Fig. 4(b)] is converted into the binary image I_b [Fig. 4(c)], which simplifies further processing.

The dilation and erosion operations are two basic morphological operations in binary image. The dilation operation is used to connect areas of similar characteristics, and the erosion operation is applied to remove bright noise points. Both the opening and closing operations are the combinations of the above two basic morphological operations. The opening operation of binary image I_b by structural elements SE is defined as Eq. (1):

$$I_b \circ SE = (I_b \ominus SE1) \oplus SE2 \tag{1}$$

thus, the opening operation of I_b by SE is to do the erosion of I_b by $SE1$, followed by a dilation of the result by $SE2$. Similarly, the closing operation of I_b by SE is defined as Eq. (2):

$$I_b \bullet SE = (I_b \oplus SE2) \ominus SE1 \tag{2}$$

which indicates that the closing operation of I_b by $SE$is the dilation of $SE1$ by $SE2$, followed by the erosion of the result by $SE1$.

Moreover, the selection of structural elements directly affects the detection performance. In this paper, a small rectangular structural element, $SE1$, is used to refine the image erosion operation as shown in Eq. (3). Then, according to the characteristics of hyperbola, the dilation operation $SE2$ for hyperbola model is selected to filter the hyperbola shape of binary images, which can filter out some shapes that are obviously different from each other as follows:

$$SE1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} SE2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \tag{3}$$

In proposed scheme, the closing operation is applied first to fill the objects with small holes, connect the adjacent objects with small gaps, and smooth their boundaries. Then, the opening operation is used to eliminate small objects and separate joint objects at slim points. The processed result after closing and opening operations is shown in Fig. 4(d).

3.2. Double cluster seeking estimate algorithm

Recently, Zhou et al. [28] proposed the OSCA algorithm to identify the point clusters with downward-opening signatures. However, the considered condition is too ideal to accommodate certain more complex situations as follows:

1) The non-target cluster also exists downward opening [Fig. 5(a)];
2) There are multiple branches in the left/right tail of the downward opening [Fig. 5(b)];
3) The two tails of the downward opening overlap at the end [Fig. 5(c)].

In view of these shortages, the DCSE algorithm is proposed to supplement and improve OSCA algorithm. After an input image is converted into a binary image, the proposed algorithm is divided into two rounds of downward opening searching, the first round is to eliminate non-target cluster areas with downward opening. In the second round, the clustering method is applied to separate the picked regions into different clusters and recognize hyperbolae.

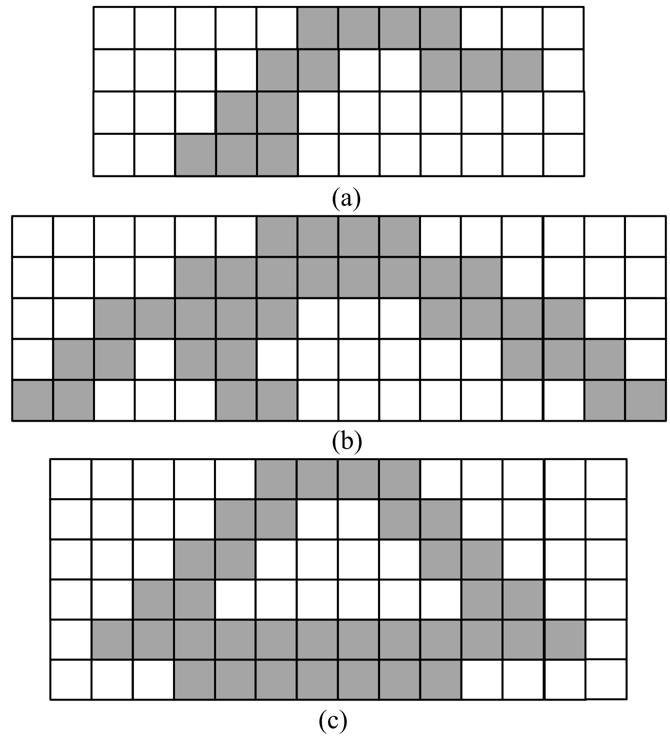

Fig. 5. Given examples of three kinds of complex situations. (a) Non-target cluster with downward opening. (b) Two branches in the left tail of the downward opening. (c) Two tails overlap at the end of downward opening.

3.2.1. Some concepts of DCSE

To explain the DCSE algorithm clearly, it is necessary to take the OSCA algorithm for reference, two concepts should be clarified first: the “downward/upward opening” and the “point segment”, which are visualized in Fig. 6. A “point segment” is more than two consecutive points in a row [Fig. 6(a)]. The preprocessed binary image is

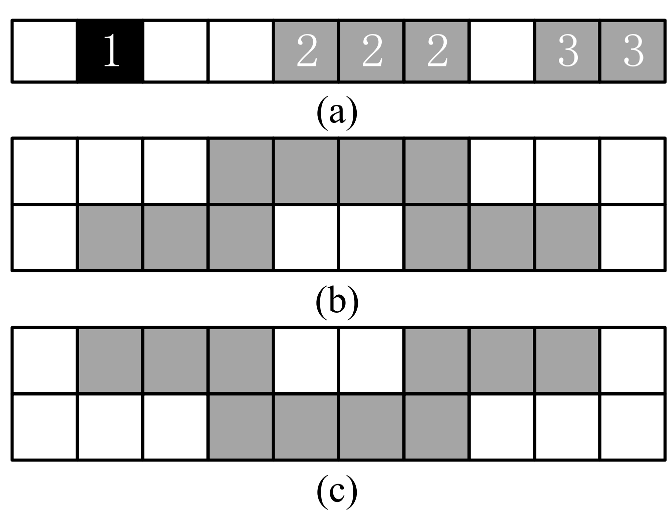

Fig. 6. (a) Non-point segmentation (tag 1) and point segmentation (tag 2 and 3). (b) Downward opening. (c) Upward opening.

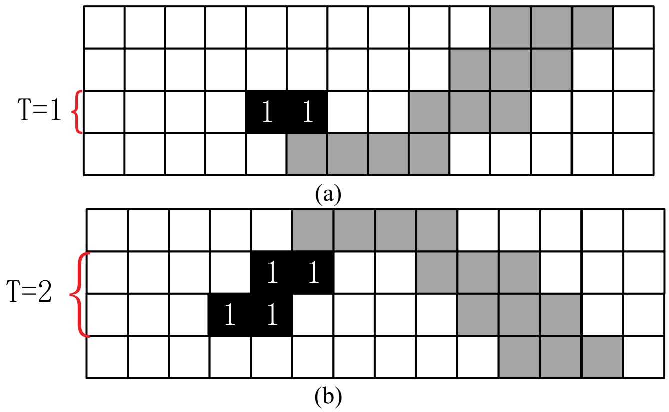

Fig. 7. Squares marked as 1 are eliminated. (a) No-target cluster with upward opening. (b) No-target cluster with downward opening.

progressively scanned from top to bottom, and thus, in some consecutive rows, point segments are obtained from which the downward opening could be defined as three connected point segments. Moreover, a point segment has horizontal overlaps with another two disconnected point segments with a gap between them in the next row, and the upper point segment is defined as starting a downward opening [Fig. 6(b)]. In the same way, the upward opening could also be defined as three connected point segments located in two consecutive rows, in which a point segment has horizontal overlaps with another two disconnected point segments with a gap between them in the previous row, and the lower point segment is defined as starting an upward opening [Fig. 6(c)].

3.2.2. Double cluster seeking estimate (DCSE)

1) Cluster seeking round 1:

Non-target cluster may also include opening as shown in Fig. 7. To address this issue, the round 1 is utilized to search all openings and eliminate irregular areas. If there exists the vertical length of a tail less than (not equal) T, the tail with point segmentations will be eliminated. The greater the value of T, the better the non-target cluster removal effect, but the characteristics of the hyperbolic cluster would also be eliminated more seriously. To maximize the retention of the hyperbolic characteristic while de-noising, in this paper, the vertical length T with the value of 3 is adopted in round 1. Fig. 7 shows two eliminated examples of no-target cluster with upward and downward openings, where the squares (marked as 1) represent the eliminated irregular area.

2) Cluster seeking round 2:

After removing the no-target clusters, the rest of clusters are searched again in the obtained output image. The position (marked as 1) of upward opening is called the "opening" and saved as Array 1 (see Fig. 8(a)). Likewise, the opening (marked as 2) of downward opening is saved as Array 2 (see Fig. 8(b)).

For a downward opening, in the process of seeking upward, if there is an opening of upward opening (tag 1) contained in Array 1, then this downward opening is regarded as a false target formed by the cross of two hyperbolae (see Fig. 9). Then the downward opening can no longer be considered as a start of point cluster, and the opening marked as 2 are removed from Array 2. Next, another new downward opening is processed.

After screening the downward opening, the point segment is scanned downward line by line. If an upward opening is found and marked, the algorithm scans adjacent point segments downward. Now,

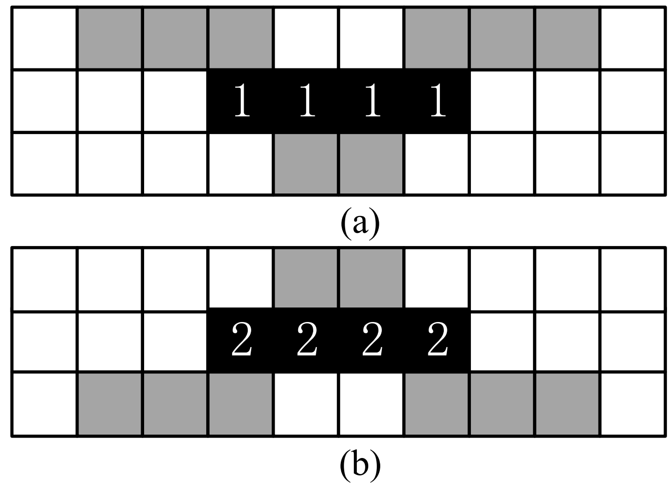

Fig. 8. Opening is saved for further processing. (a) Opening of upward opening (tag 1). (b) Opening of downward opening (tag 2).

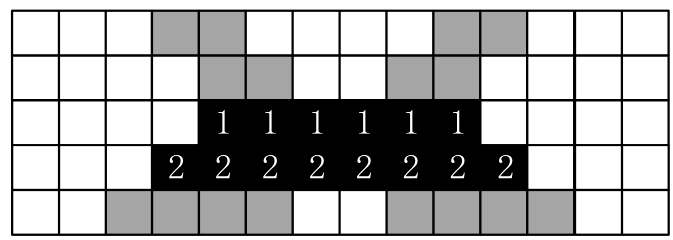

Fig. 9. A false target formed by the cross of two hyperbolae.

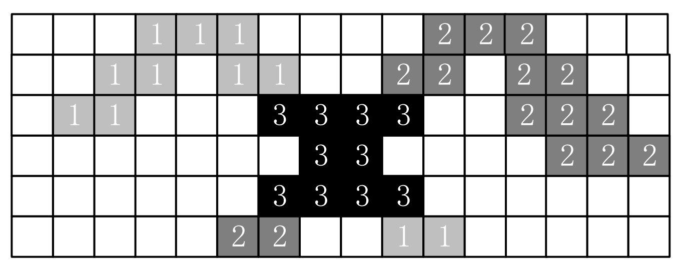

Fig. 10. The point segment that starts the upward opening would be passed down until a downward opening is found. The point segments in the right tail would be added into the cluster of the upper-left point segments (tag 1), while the point segments in the left tail would be added into the cluster of the upper-left point segments (tag 2), and the squares tagged as 3 between the upward and downward openings would be deleted.

there exist two situations that are going to be discussed:

1) One situation is that a downward opening is found as shown in Fig. 10, in such case, the point segments in the right tail would be added into the cluster of the upper-left point segments (tag 1), while the point segments in the left tail would be added into the cluster of the upper-right point segments (tag 2), and the squares tagged as 3 between the upward and downward openings would be deleted.

2) The other situation is that there is no downward opening below, which indicates that two hyperbolae intersect at the end of their tails. And thus, when the point segment with no adjacent ones below it is found, all the point segments below the upward opening (including upward opening) would be deleted. An ideal case shown in Fig. 11, the squares marked as 2 and 3 represent different clusters, which intersect at the end of their tails. The overlapped squares (tag 1) below the upward opening would be deleted.

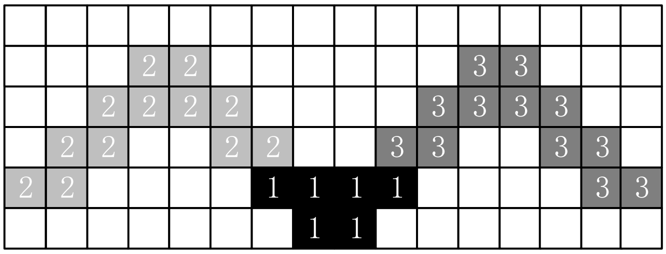

Fig. 11. Two clusters intersect at the end of their tails. Squares below the upward opening (tag 1) would be eliminated. Other squares (tag 2 and 3) represent different clusters.

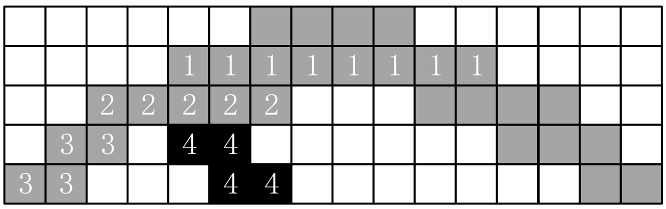

Fig. 12. Multi branches overlap with previous point segment in left tail of downward opening. The opening (tag 1), and left tail (tag 3) and right tail (tag 4) overlap with previous line (tag 2), where left tail is retained, while right tail is eliminated.

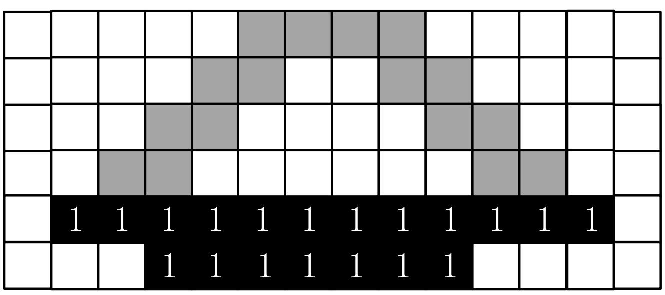

Fig. 13. Left tail intersects with right tail in the same hyperbola. And the intersected area (tag 1) is deleted.

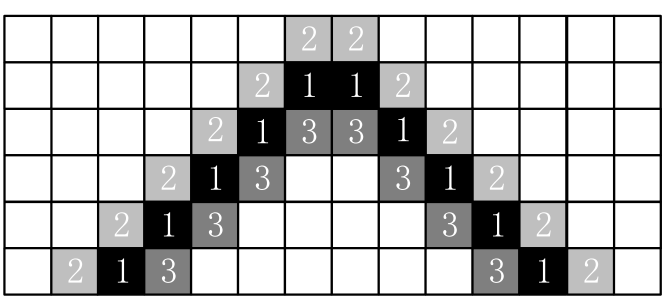

Fig. 14. An ideal case of obtaining fitting points (tag 1) via CTFP method.

In addition, during the seeking downward, we discussed two special situations: 1) For left tail of a point cluster, if multiple point segment in the same line overlap with adjacent previous one, in those the left-most tail is retained and others are ignored. In the same way, the right-most tail is assigned to lower right one of the downward opening. An example shown in Fig. 12, the opening of a downward opening (tag 1) is found first, by scanning the point segment downward in left tail of downward opening, there are two branches that are found to overlap with the previous point segment (tag 2), thus the left tail (tag 3) is retained and the right tail (tag 4) would be deleted. 2) The left tail intersects with the right tail in the same cluster shown in Fig. 13, in which it only needs to remove the intersected area (tag as 1).

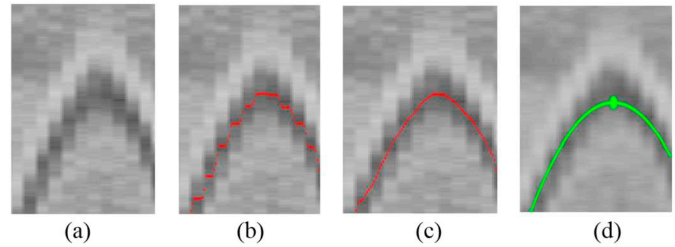

Fig. 15. Hyperbola fitting based on CTFP method. (a) GPR B-scan image. (b) Obtained initial input points (red dots). (c) Obtained smoothed fitting points (red dots). (d) Hyperbola fitting result with peak (green dot). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Hyperbola localization

4.1. Column-based transverse filter points

In each target region, hyperbolic curve fitting enables exact localization of the target through provided a series of fitting points. For the obtained formation of hyperbola signatures after DCSE method, we propose the column-based transverse filter points (CTFP) method to extract automatically a set of mean-position points in each column.

First, each hyperbolic cluster is scanned column by column to find the upper endpoint $P_1 = (x_0, y_1)$ and lower endpoint $P_2 = (x_0, y_2)$ of each column. The mean value of the upper and lower endpoints is obtained, that is to say, the value of y-coordinates is obtained, and x-coordinate remains same. And then each candidate fitting point $P_t = \left(x_0, \frac{y_1 + y_2}{2}\right)$ of each column can be obtained. In ideal case as in Fig. 14, we obtain fitting points via the proposed CTFP method, where P_t, P_1, and P_2 are tagged as 1, 2, 3, respectively.

The input points used for hyperbola fitting could contain outliers due to noise and unwanted subsurface reflections. In order to make curve fitting smoother, a lower pass filter is applied to smooth the alternative fitting point $P_n = (x, y)$. Based on the statistical law, the lower pass filter regards the continuous sampling data as a queue whose length is fixed to N. After the new measurement, the first data of the above queue is removed, and the remaining $(N - 1)$ data is moved forward successively, and the new sampling data is inserted as the tail of the new queue. The queue is then arithmetic, and we can obtain the final validated point $P_n' = (x', y')$ by replacing P_n with the average value of the adjacent N potential fitting points:

$$\mathrm{x(n)}' = \mathrm{x(n)} \tag{4}$$

$$y(n)' = \frac{1}{N}\sum_{i=1}^{N} y(n - i + 1) \tag{5}$$

As shown in Fig. 15, hyperbola is fitted using CTFP method in presence of few outliers. By using CTFP to select the fitting points, hyperbola fitting improves and peak (green dot) is found correctly. In this paper, the window width is set to 10.

4.2. Hyperbolic curve fitting

From the formation of hyperbola signatures from the B-scans, the obtained mean-position fitting points in each column by CTFP form a hyperbola. These points are used as input points for fitting a north-south opening hyperbola as Eq. (6):

$$\frac{(y - k)^2}{b^2} - \frac{(x - h)^2}{a^2} = 1 \tag{6}$$

where (h, k) is the center of hyperbola, a is length of semi-major axis, b is length of semi-minor axis. Random sample consensus (RANSAC) [41] is an iterative method that can be used for robust fitting in presence of

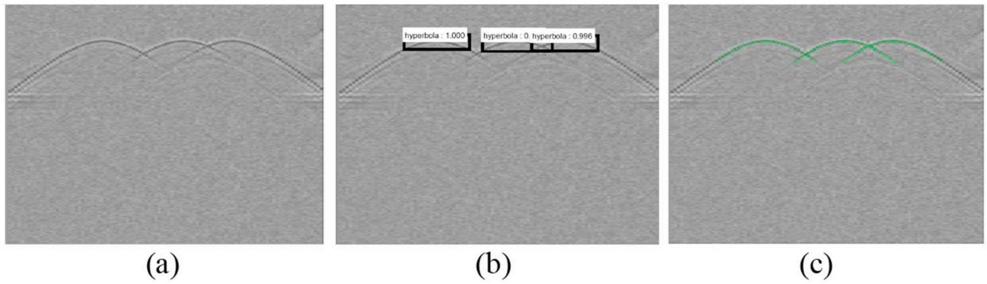

Fig. 16. The proposed scheme conducts on first synthetic B-scan data. (a) B-scan image. (b) Detecting results of Faster R-CNN. (c) Fitting results.

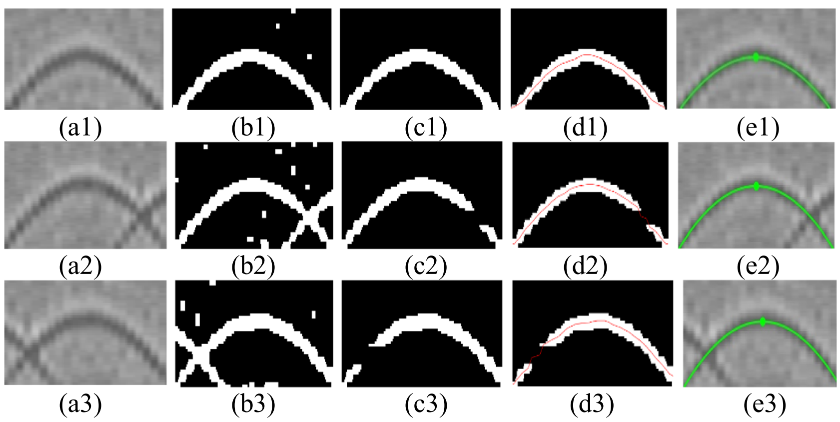

Fig. 17. The intermediate results on the first synthetic experiment. (a1–a3) Extracted detecting results of Fig. 16(b). (b1–b3) Preprocessed binary image. (c1–c3) Obtained result of DCSE. (d1–d3) Extracted fitting points after CTFP (red dots). (f) Fitting results with peak (green dot). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

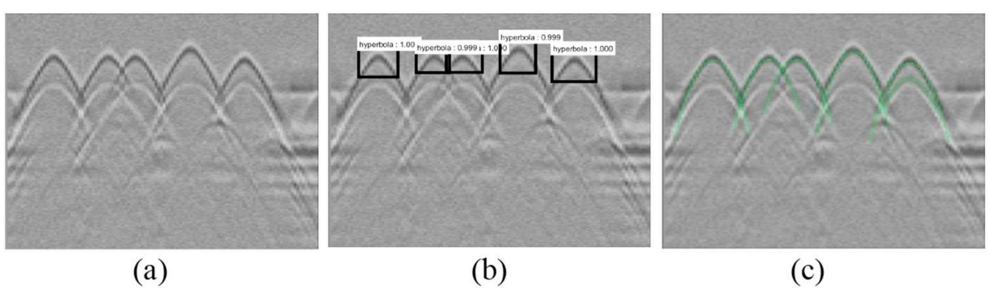

Fig. 18. The proposed scheme conducts on the second synthetic B-scan data set. (a) B-scan image. (b) Detecting results of Faster R-CNN. (c) Fitting results.

outliers. Using RANSAC, hyperbola fitting is corrected as shown in Fig. 15(d).

5. Experiments and discussion

5.1. Experimental setup

All the programs are implemented using Detectron (https://github.com/facebookresearch/Detectron) library. Detectron is an open source target detection platform released by Facebook AI research institute (FAIR), which makes us build model and application quickly with the most advanced deep-learning target detection technology. The code was run on a server equipped with an Intel Core i7-7700 of 3.6GHz and NVIDIA GTX 1060 GPU.

For implementation, we adopt the Caffe2 framework (https://github.com/caffe2/caffe2) that is the updated version of Caffe [42] to train our Faster R-CNN models. For the RPN classification part, a selected region proposal is regarded as a hyperbola if the classification confidence score is greater than 0.6 in our experiments, and non-hyperbola otherwise.

In fact, to effectively train a CNN, datasets of thousands or even millions of images are typically used [43]. It is a big issue to acquire a huge number of labeled GPR B-scans. However, one of the strong aspects of the Faster R-CNN framework is the possibility of being trained on synthetic images, still being able to work when deployed on on-site GPR acquisitions. In this work, we evaluate the proposed hyperbola detection solution on our own GPR training set, which initially has a total of 838 images, including synthetic data set (748 images) and on-site data set (90 images). After applying DA (e.g., horizontal and vertical stretching and compression, horizontal flipping, image enhancement), we obtain a total of 5866 images composed of 5236 synthetic images and 630 on-site images. We divide 50% of the hyperbola dataset as training set and remaining 50% as testing set. To verify the importance of DA on Faster R-CNN, the performance comparisons are reported on our hyperbola data set as shown in Table 1. We can clearly

Table 1
The accuracy comparison of Faster R-CNN with/without DA.

Iterations	Accuracy (%)	
	Without DA	With DA
2000	87.23	91.15
3000	86.80	**92.13**
4000	87.32	91.02
5000	87.61	90.37

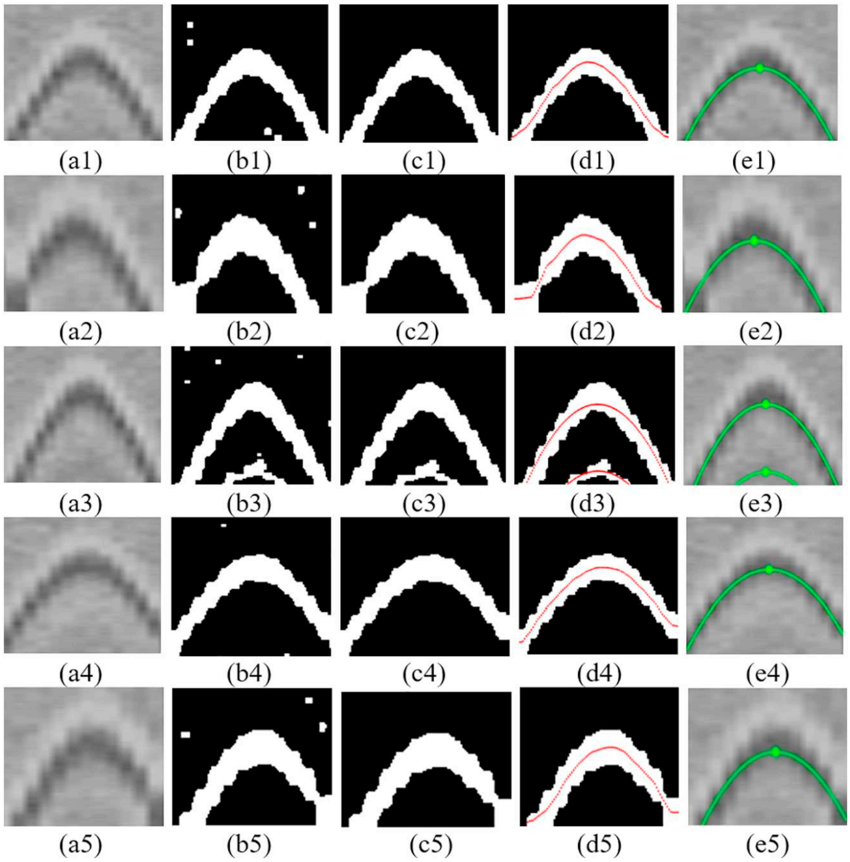

Fig. 19. The intermediate results on the second synthetic experiment. (a1–a5) Extracted detecting results after Fig. 18(b). (b1–b5) Preprocessed binary image. (c1–c5) Obtained result of DCSE. (d1–d5) Extracted fitting points (red dots) after CTFP. (e1–e5) Fitting results on extracted target region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

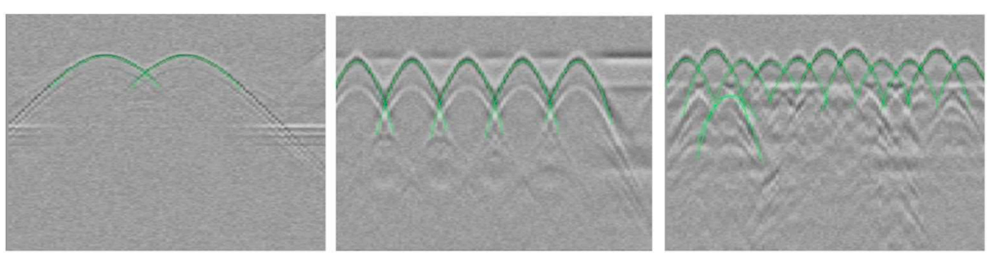

Fig. 20. Some experimental results on synthetic data.

Fig. 21. In our experiment, the LTD-2200 GPR with GC400-MHz antenna is utilized to obtain the real GPR B-scan images.

see that the overall accuracy of Faster R-CNN with DA is better than that without DA. In specific, the former just takes 3000 iterations to reach an accuracy of 92.13%. More details of the experimental results are given in the following sub-Sections.

5.2. Experiments on synthetic data sets

The synthetic GPR B-scan images are generated using GprMax simulation software [43]. We generated synthetic images of different ground compositions (i.e., different sands, clays, and compositions) containing objects of different shapes (i.e., boxes, spheres and cylinders) with diverse dielectric constants. In order to make our experiment more convincing on the synthetic data set, each synthetic image was added noise with a signal-to-noise ratio of 10 dB to be closer to on-site image. In our experiments, the reflectance of the ground surface has been eliminated in advance.

First, we applied the proposed scheme on synthetic data sets shown in Fig. 16. There are intersections between the hyperbolae in the generated B-scan image [Fig. 16(a)]. There exist three detecting results containing hyperbola signature based on Faster R-CNN method in Fig. 16(b). In addition to rectangular boxes, each box has a confidence

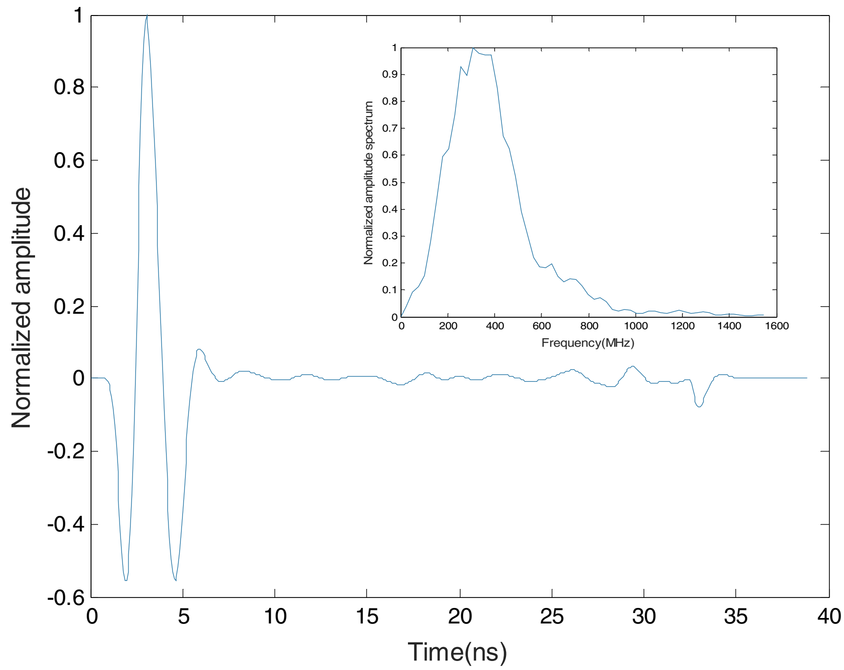

Fig. 22. Direct coupled signal and it's spectrum.

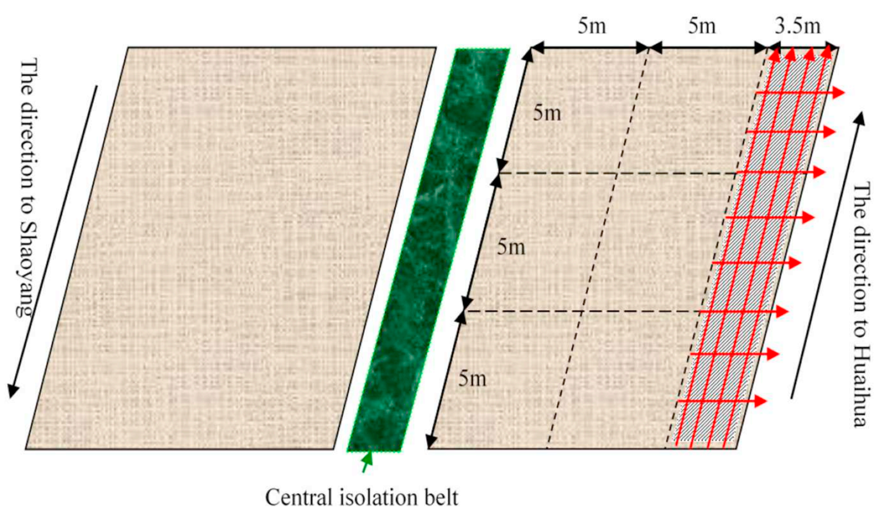

Fig. 23. The sketch of on-site experimental environment for Shaoyang-Huaihua highway. The red arrow shows the moving direction of the GPR antenna.

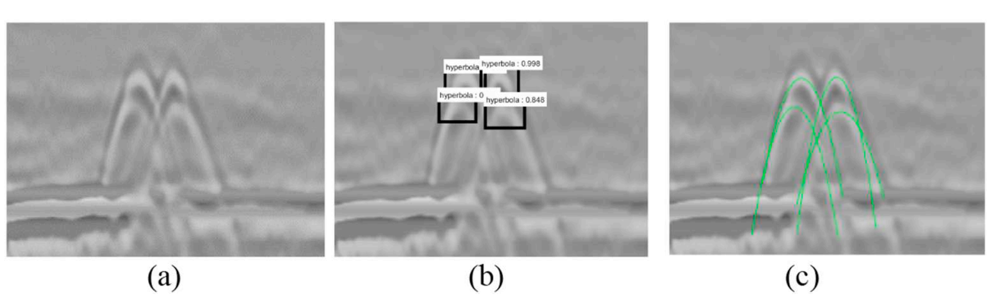

Fig. 24. The proposed scheme conducts on the first on-site B-scan image. (a) B-scan image. (b) Detecting results. (c) Fitting results.

in the upper right corner that indicates the probability that the target is a hyperbola. Following the fitting results in Fig. 16(c), which does not leave out any hyperbola target. Fig. 17 supplements the following procedures of each detecting region after Faster R-CNN in detail. The three detecting results are extracted solely, which are sorted from top to bottom in descending order of confidence: 1.000, 0.998, 0.995 [Fig. 17(a1–a3)]. Some noise appears on the image, which have not been completely eliminated in the preprocessing step [Fig. 17(b1–b3)]. Fig. 17(c1–c3) show the obtained clustering results by DCSE. The input fitting points are extracted and smoothed automatically by CTFP [Fig. 17(d1–d3)], in which each point clusters with hyperbolic signature is identified and fitted [Fig. 17(e1–e3)].

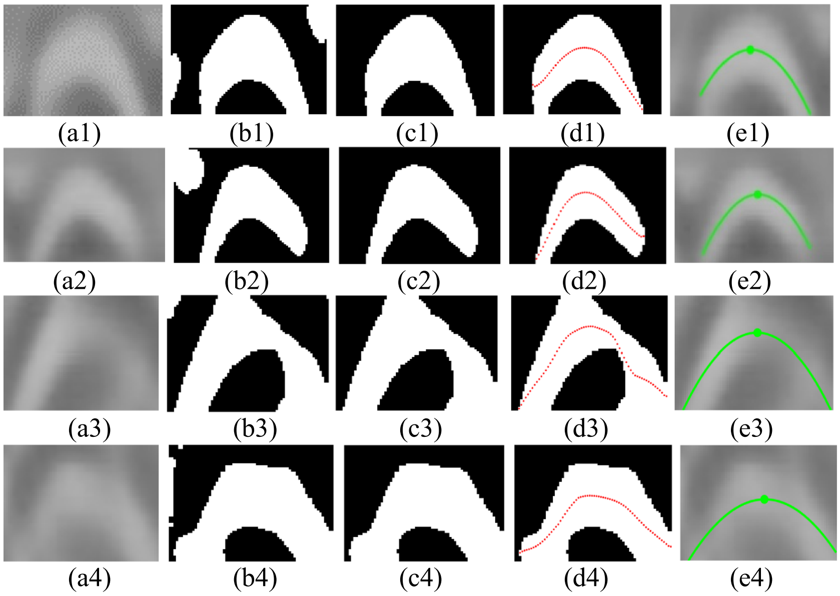

Fig. 25. The intermediate results in the processing of the first on-site experiment. (a1–a4) Extracted detecting results after Fig. 24(b). (b1–b4) Preprocessed binary image. (c1–c4) Obtained result of DCSE. (d1–d4) Obtained fitting points (red dots) after CTFP. (e1–e4) Fitting results on extracted target images. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

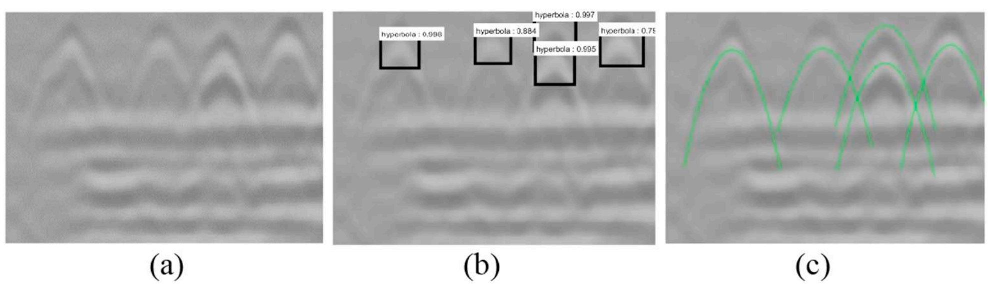

Fig. 26. The proposed scheme conducts on the second on-site B-scan image. (a) B-scan image. (b) Detecting results. (c) Fitting results.

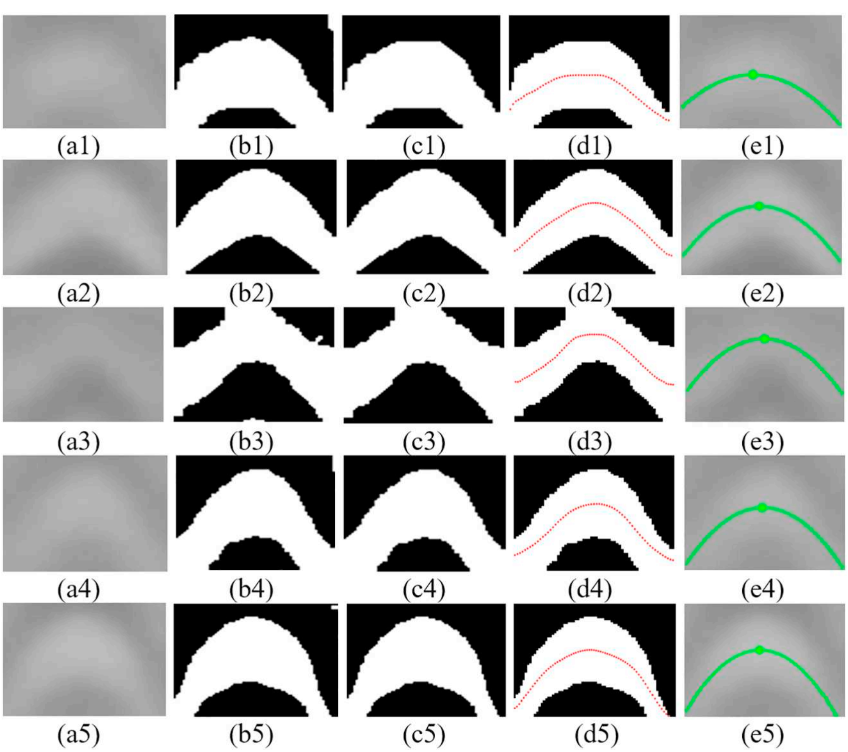

Fig. 27. The intermediate results on the second on-site experiment. (a1–a5) Extracted detecting results after Fig. 26(b). (b1–b5) Preprocessed binary image. (c1–c5) Obtained result of DCSE. (d1–d5) Extracted fitting points (red dots) after CTFP. (e1–e5) Fitting results with hyperbolic peak (green dot). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The second synthetic experiment is shown in Fig. 18, some complex intersections, cavity reflection and echo effects appear on the B-scan image [Fig. 18(a)]. The experimental result demonstrates that Faster R-CNN could return rectangular regions that might contain hyperbolae [Fig. 18(b)], followed by the hyperbolic curve fitting [Fig. 18(c)]. The five detecting results by Faster R-CNN are extracted solely, which are sorted in descending order of confidence: 1.000, 1.000, 1.000, 0.999, 0.999 [Fig. 19(a1–a5)]. After preprocessing, the B-scan image is transformed into the binary image [Fig. 19(b1–b5)], and some irregular shapes and derived hyperbolic shapes appear on the binary image due to the intersections and echo effects. DCSE is utilized to eliminate these noises visualized in Fig. 19(c1–c5). The input sample points are extracted and smoothed automatically by CTFP method [Fig. 19(d1–d5)]. As we can see, the only disputed case appears in the second synthetic experiment, in which six clusters are obtained by OSCA while there are only five buried objects, and the extra one hyperbolic signature is derivative reflections of the target. Although there will be a redundant target, the results of the proposed scheme are acceptable, since all the five hyperbolic signatures are found effectively. The following fitting results are shown in Fig. 19(e1-e5). More experimental results on synthetic data sets are displayed in Fig. 20. It can be seen that all the hyperbolic signatures are detected.

5.3. Experiments on on-site data sets

During the scheme validating procedure, it is necessary to consider on-site data from GPR acquisitions as experimental data sets. Specifically, our on-site data sets are collected using LTD-2200 GPR with GC400-MHz antenna [Fig. 21]. This antenna pair has internal bow-tie structure with T/R offset of 16 cm. Its external size of $32 \times 32 \times 21\ \text{cm}^3$ including protective shell. We recorded T/R direct coupled signal by reversing the antenna pair to the sky. The direct coupled signal and its amplitude spectrum are shown in Fig. 22. We used this GPR equipment to detect rebars buried under the Shaoyang-Huaihua highway, located at Hunan Province. And the detection scanning scenario is shown in Fig. 23.

First, we conduct the first on-site experiment with the proposed method in Figs. 24 and 25. Be different with the synthetic data, the on-site data is along with much noisy. For original gray B-scan image [Fig. 24(a)], Faster R-CNN captures four regions including all the expected hyperbolae [Fig. 24(b)] for hyperbola fitting [Fig. 24(c)]. The details of obtained detecting results are visualized in Fig. 25. By applying the preprocessing step, the picked region presented in Fig. 25(a1–a4) is transformed into the binary image containing the target hyperbola and some irregular shapes [Fig. 25(b1–b4)]. The DCSE is then utilized to process the binary image [Fig. 25(c1–c4)]. As CTFP method, it is used to acquire a series of fitting points [Fig. 25(d1–d4)]. Naturally, each fitting curve of gray target region is easy to obtain as shown in Fig. 25(e1–e4).

With more complicated noises and the horizontal underground media, the second on-site data sets are generated from four buried rebars to verify the efficiency and accuracy of our scheme. As shown in Fig. 26, Faster R-CNN algorithm works effectively to pick all the expected targets for hyperbolic fitting. As we can see, there is a redundant detection obtained by Faster R-CNN, which is a derivative reflection of the target. Although there will be a redundant target, the results of the proposed scheme are acceptable, since all the four hyperbolic signatures are found effectively. Fig. 27 shows the intermediate results in the processing. The picked B-scan regions and the preprocessed binary images are visualized in Fig. 27(a1–a5) and (b1–b5), respectively. After preprocessing, the image becomes much smoother with part of the cracks and discrete noisy points being eliminated, which is helpful for the subsequent process. After applying DCSE, there are only one cluster left of each target region [Fig. 27(c1–c5)]. The results of CTFP are shown in Fig. 27(d1–d5), in which only one cluster is fitted [Fig. 27(e1–e5)]. In both two on-site experiments, all the correct targets are identified.

Table 2
Performance of the proposed scheme for four highway areas.

Highway (No.)	Ground-truth number of rebar peaks	TP	FP	FN	Recall (%)	Precision (%)
1	156	152	7	4	97.44	95.60
2	233	230	6	3	98.71	97.46
3	409	398	22	11	97.31	94.76
4	218	213	10	5	97.71	95.52
Overall	1016	993	45	23	**97.74**	**95.66**

In addition, in order to verify effectiveness of the proposed scheme in engineering application, we add several additional on-site experiments derived from surveys of four highway areas. This GPR data is also collected from Shaoyang-Huaihua highway, in which the horizontal scanning lengths are 165 m, 285 m, 345 m and 255 m, respectively. The rebar targets are distributed at an underground depth of about 20 cm. Different from the continuous and uniform distribution of rebars in bridge deck [11], the rebars in our data set only distribute at the junction of the adjacent decks on the highway, and are used to reinforce highway decks. Our own on-site data set is more complex and difficult to obtain and handle.

The results of the automatic rebar localization for each highway area are summarized in Table 2, which displays clearly the number of true positive (TP), false positive (FP), and false negative (FN) targets, respectively. For a comprehensive evaluation, two measures of performance are utilized, as defined by precision (P) and recall (R). It can be learned that the overall recall and precision are observed to be 97.74% and 95.66%. The smallest recall and precision are 97.31% and 94.76% for the 3th road. Our scheme could pick most of hyperbolic targets. It demonstrates the necessity of having our scheme on on-site data.

5.4. Comparisons between the proposed scheme and other methods

In order to better verify the efficiency of the proposed scheme, this section presents the comparison results from different methods and discussed below.

1) Compared detecting results between Faster R-CNN and VJ method

For the detection of the candidate hyperbola regions, we first compare the Faster R-CNN with the VJ-based detector [24] on a whole gray B-scan image. In our scheme, by applying Faster R-CNN object detection method, the hyperbolic regions are identified from the original B-scan image, and the confidence is also marked on the top right corner of the rectangle. The results of the VJ method are shown in Fig. 28, compared with the detecting results of Faster R-CNN in Figs. 24(b) and 26(b), hyperbola could be identified among the two methods without missing any target, while there exists redundant detection boxes (blue) for the VJ method, that is because the haar-like features used in the VJ method cannot fully represent hyperbola features. In addition, two additional on-site experiments are presented in Fig. 29. Be different from the obtained result by the Faster R-CNN [Fig. 29(a)], the VJ method identities hyperbolic curves with a missing target (red) shown in Fig. 29(b). In Fig. 29(c) and (d), both two methods perform well without missing any target, but the VJ method still has a redundant detection result (blue). The compared results indicate that our method can be applied on complicated data sets effectively, further, the accuracy of Faster R-CNN is higher than VJ method.

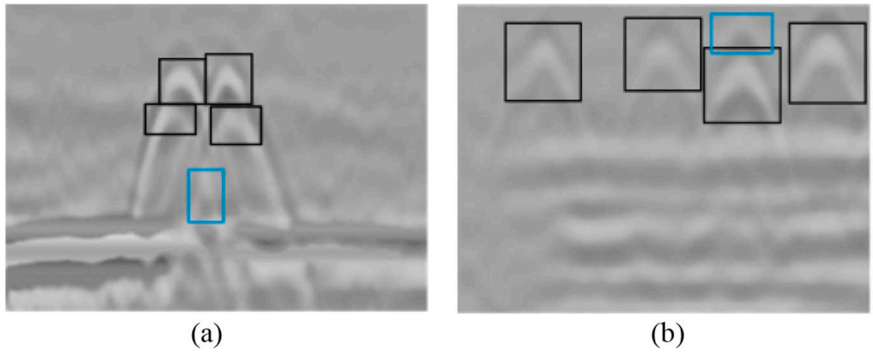

Fig. 28. Obtained results of the VJ method with true target (black) and redundant target (blue). (a) Obtained result on the first on-site data set. (b) Results on the second on-site data sets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

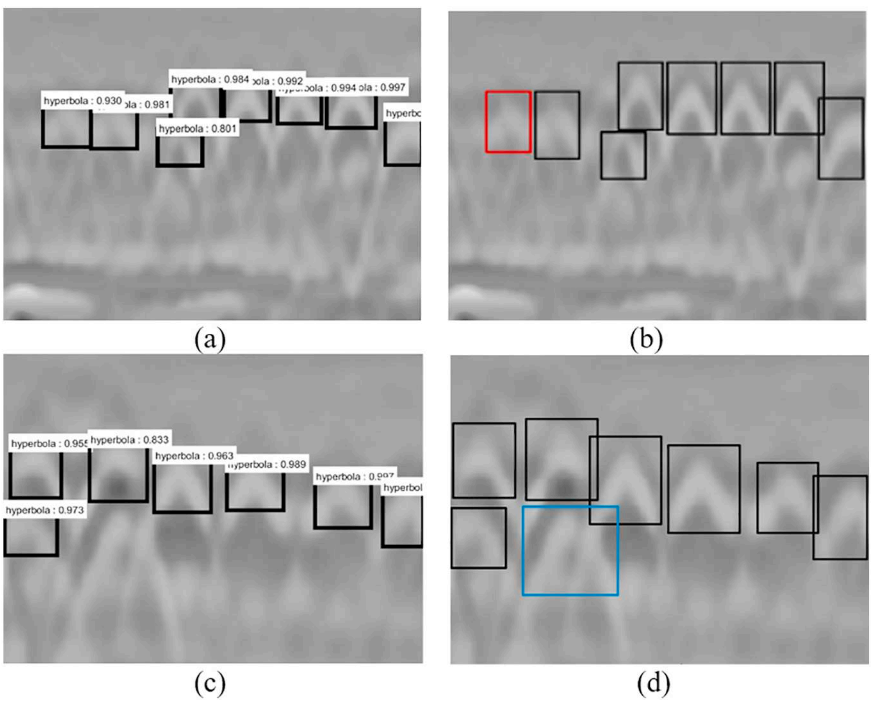

Fig. 29. Results of the Faster R-CNN compared with VJ method on additional on-site data sets with true target (black), redundant target (blue) and missed target (red). (a) and (c) Obtained result by Faster R-CNN. (b) and (d) Results by VJ method. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2) Clustering results on whole image by our scheme

Another experiment is designed to verify the importance of utilizing the Faster R-CNN. Here we implement the proposed DCSE clustering method directly on whole image without considering Faster R-CNN method. The clustering results on the first and second synthetic data sets are shown in Fig. 30, and on first and second on-site data sets are presented in Fig. 31. It can be found that, when without Faster R-CNN, the whole B-scan might contain hyperbolae, more lines, and discrete points, which cause significantly the increased number of missing detection clusters. But the proposed scheme with Faster R-CNN can achieve majority of target extraction and effectively narrow down background noise.

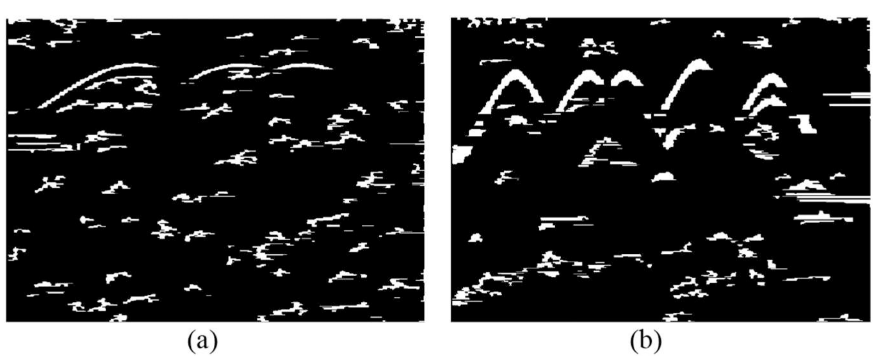

Fig. 30. Clustering results on whole synthetic image by DCSE without Faster R-CNN detecting. (a) Obtained result on the first synthetic experiment. (b) Results on the second synthetic experiment.

Fig. 31. Clustering results on whole on-site image by DCSE without Faster R-CNN detecting. (a) Results on the first on-site data set. (b) Results on the second on-site data set.

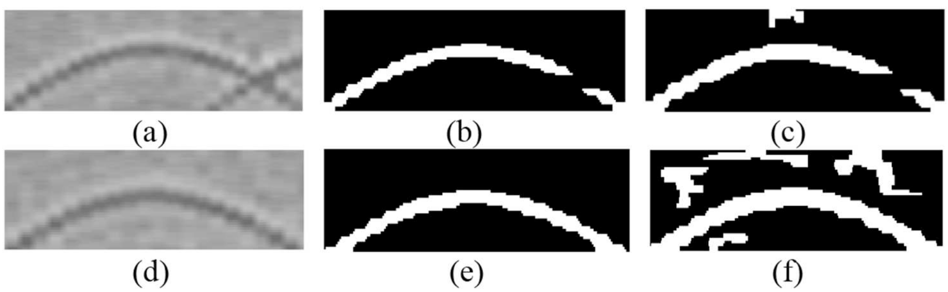

Fig. 32. Compared clustering results of DCSE and OSCA. (a) and (d) Original images. (b) and (e) Obtained result by DCSE. (c) and (f) Obtained results on by OSCA.

3) Compared clustering results between DCSE and OSCA methods

The comparisons of clustering result between the proposed DCSE and OSCA algorithm [28] are conducted. Both the methods could perform unsupervised clustering on processed binary B-scan images and separate intersected hyperbolic curves. After the same preprocessing method and detecting step, both the two cluster algorithms are conducted on picked candidate region, and the compared clustering results are shown in Fig. 32. It can be learned that the processing time of DCSE and OSCA are almost the same, but the performance of DCSE is better than OSCA. Since there exists a more complex situation that the non-target cluster also exists downward opening, where OSCA conducts clustering by the opening information cannot deal with this situation, while DCSE scans image twice and set the threshold for the opening information to remove this kind of cluster.

6. Conclusions

This paper describes our proposed scheme for interpreting GPR B-scan images, which consists of Faster R-CNN, preprocessing method, DCSE, and CTFP method. The gray object regions containing expected hyperbola targets are first extracted by Faster R-CNN from GPR B-scan image. Then, the obtained point clusters in target region could be segmented and classified by using DCSE method, followed by CTFP that further obtains hyperbola fitting points automatically. Finally, a series of fitting points are utilized to fit the identified point clusters, and thus, the peak of the object could be obtained. To solve the problem of lacking on-site data, we adopt the DA strategy to increase the volume and variety of the training data, so as to improve the performance of Faster R-CNN on GPR data. Being different from previous clustering methods, DCSE is concerned with more complicate cases including non-target cluster with downward opening, multiple branches in the left/right tail of the downward opening and two tails of the downward opening overlap at the end, by which nonhyperbolic point clusters are eliminated and some hyperbolae that cross and intersect with others are also separated into different clusters. By computing the median of the boundary endpoint of hyperbolic point cluster, the proposed CTFP is utilized to obtain fitting points which then are filtered by a low pass mean filter. The obtained fitting points can be used in further applications such as fitting hyperbola and estimating the peak of target.

For traditional GPR target detection, there exists some phenomena such as incomplete extraction of target signatures, false alarm and leakage alarm. Comparison with traditional GPR shows that this new scheme is more accurate and robust in terms of real-time detection and localization of targets in the experiments. In the future work, we will focus on latest deep learning framework aiming at object detection, achieving greater improvements in accuracy and speed.

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